Retrospective estimation of mortality patterns in Somalia

Report 3 (December 2023)

STATISTICAL ANNEX 3:
App Development for Forecasting Scenarios

14 December 2023

Supported by:

Table of contents

Introduction 3
Data, Temporal Patterns and Correlations 3
Methods 5
Default Scenarios: Direct Mean and Quantiles 5
User Specified Scenarios: Modified Predictors with Correlation Matrix 5
RShiny App 5
Input Parameters 6
Specificities & Visualization 7
Results 8
Limitations 9
References 10

Introduction

In our study, we used a retrospective prediction approach to estimate the crude mortality rate in Somalia, using a generalised linear mixed model (GLMM) method [5]. This method involves the incorporation of predictor variables, in particular measles incidence (per 100,000 population), malaria incidence (per 100,000 population), malnutrition incidence (per 100,000 population) and the combined drought index, collectively referred to as the model predictors. These predictors were available at district level throughout the study period and played a key role in our analysis.

To explore how the crude mortality rate may change over a short five-month forecast period, we developed an app which runs a number of different forecast scenarios; a 'default forecast scenario', and 'user-specified scenarios'. The 'default scenario' assumes constancy in the values of the predictor variables over time, maintaining the same values as those observed over the previous six months. In contrast, the 'user-specified forecast scenario' allows for customised adjustments based on user-defined parameters. In both scenarios, the process involves forecasting the model's predictor variables for the next five months, which are then fed into the GLMM to forecast the crude mortality rate. Figure 1 shows how the the retrospective analysis and forecasting relate to each other.

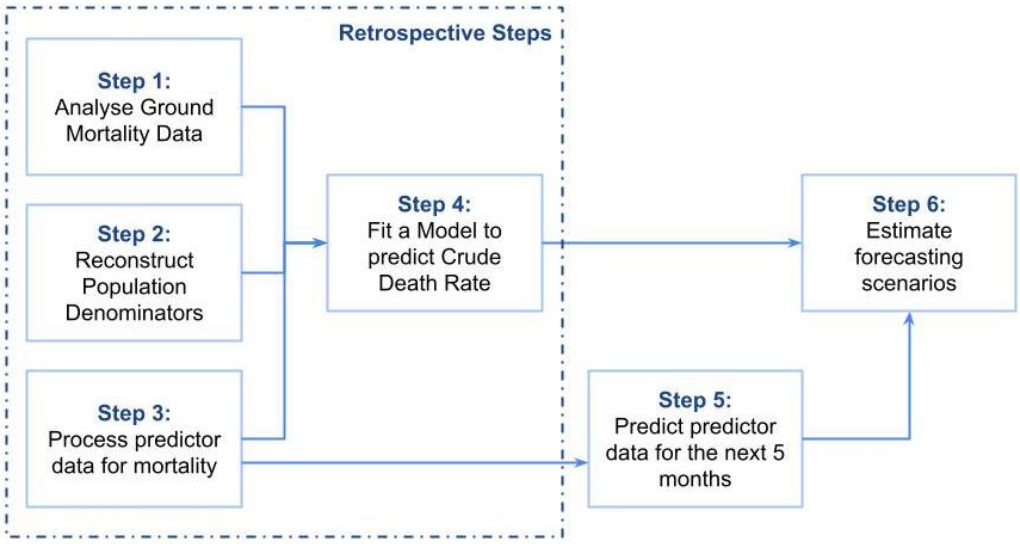

Figure 1. Schematic of forecasting scenarios framework.

Data, Temporal Patterns and Correlations

Our generalised linear mixed model (GLMM) is structured around four main predictors: Incidence of measles (per 100,000 population), Incidence of malaria (per 100,000 population), Incidence of malnutrition (per 100,000 population) and Combined Drought Index. Although the full model uses more than just these four predictors, with a full set of additional predictors described in Table 1 in Appendix 1. The data driving the model comes from a variety of sources, including WHO, FSNAU and UNICEF.

Traditionally, time series forecasting methods are used in this type of predictive analysis [1, 2]. However, we cannot use those methods due to the absence of seasonality or discernible temporal patterns in the various predictors, as Figure 2 clearly shows.

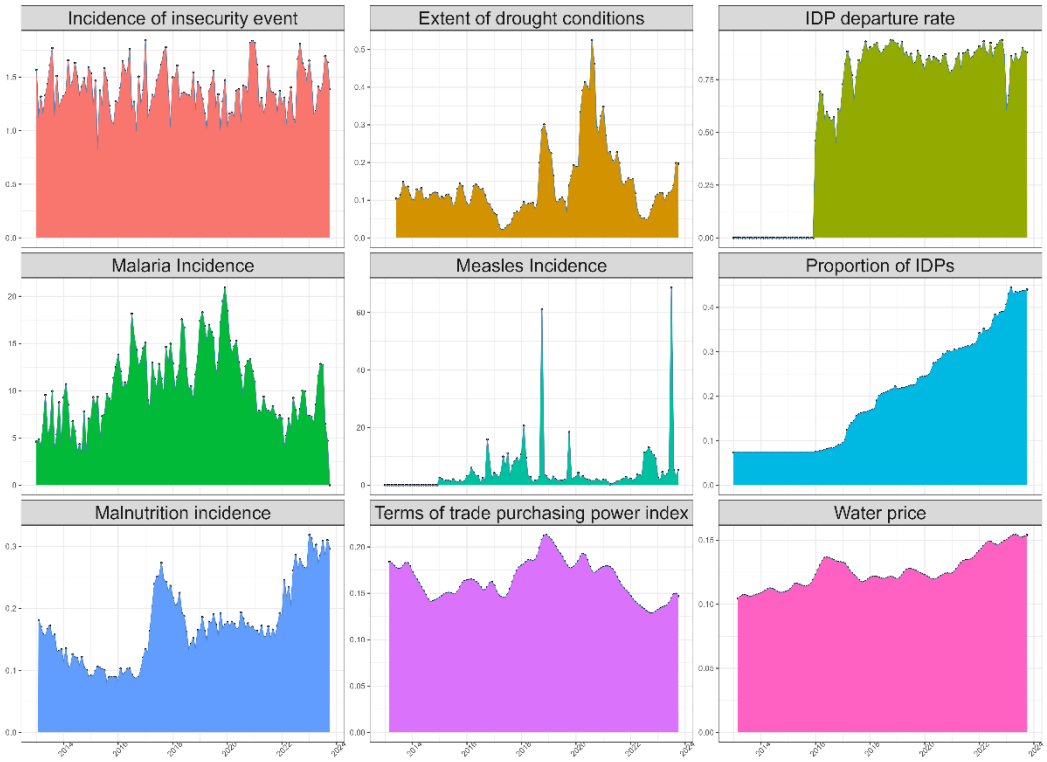

Figure 2. Visualisation of each predictor as a function of time.

We therefore try to include the areas of causality and correlation. By drawing on both on-the-ground knowledge and statistical analysis [4], as shown in Figure 3, we uncover significant information about the interaction of predictor variables.

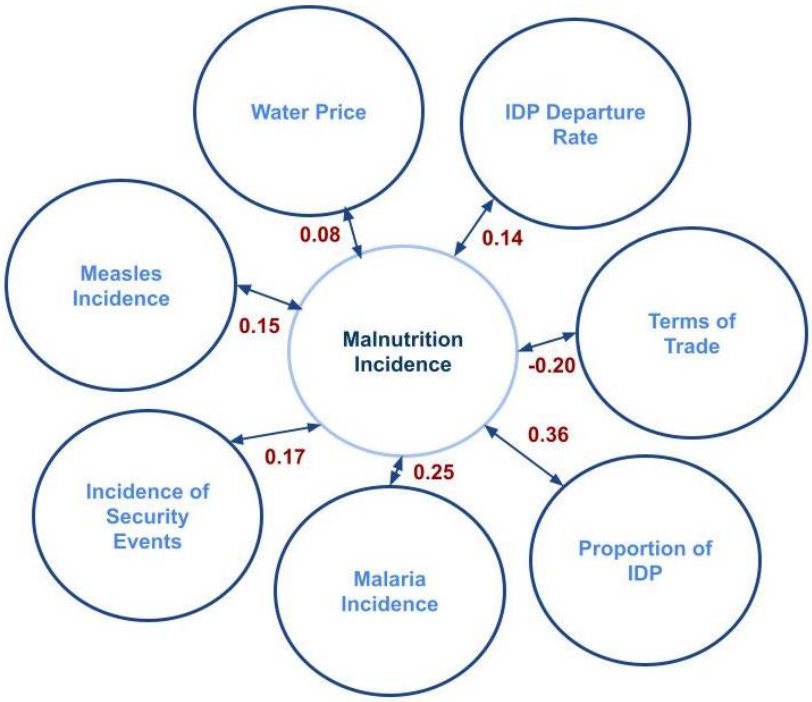

Figure 3. Example of links between one model predictor (here Malnutrition Incidence) and other predictors. Red values correspond to the correlation coefficients.

In addition, some predictors are time-lagged. For example, the combined drought index has a time lag of 5 months. Incorporating this lag gives access to the actual values of the combined drought index for the following five months. Consequently, lagged predictor variables can be used directly with their actual values, avoiding the need to forecast over the five-month period. Conversely, other predictors require forecasts over the same period.

Methods

Default Scenarios: Direct Mean and Quantiles

In the default forecast scenario method, three distinct scenarios have been meticulously developed to anticipate potential variations in predictive values over a five-month forecast period:

- a **central, 'static' scenario**: assumed conditions remain unchanged, calculated as the rowing median value of each predictor variable per district. This scenario is based on the last five months of observation.
- a **'pessimistic' scenario:** assumed conditions remain unchanged, calculated as the rowing upper quantile value of each predictor variable per district. This scenario is based on the last five months of observation.
- an **'optimistic' scenario:** assumed conditions remain unchanged, calculated as the rowing lowest quantile value of each predictor variable per district. This scenario is based on the last five months of observation.

User Specified Scenarios: Modified Predictors with Correlation Matrix

In the initial strategy, scenario are simple and pre-set. However, given the dynamic nature of real-life situations, particularly in times of crisis, users need to be able to explore possible effects of different scenarios. This adaptability is essential to accurately reflect conditions in the field. These scenarios give users the option of applying percentage increases or decreases to predictive values. As a result, at the end of the forecast period, the predictive values will be multiplied by the specified percentage, deviating from a constant trend over time. For example, if an increase of 50% is defined for "malnutrition incident", this will result in a monthly increase of 10%, for a total increase of 50% in the last month of the forecast period.

In addition, this method uses the correlation matrix defined in the generation of the model allowing users to not just alter the main predictors, but for the interaction between them to be included. Consider a scenario in which A is a model predictor, and B and C are two other predictors linked by B=b*A and C=c*A, with b and c correlation coefficients. If B is increased by x and C by y, then A will be increased by b*x*c*y. Thanks to these relationships, various alternative prediction scenarios can be defined.

RShiny App

To facilitate user interaction and streamline the process of defining percentage modifications and generating various scenarios, we have developed an App with specific features. This App has been developed through RShiny [3] and offers an intuitive interface for manipulating visualization parameters and modifying predictors. It provides a user-friendly platform to explore and generate diverse scenarios based on the specified input parameters.

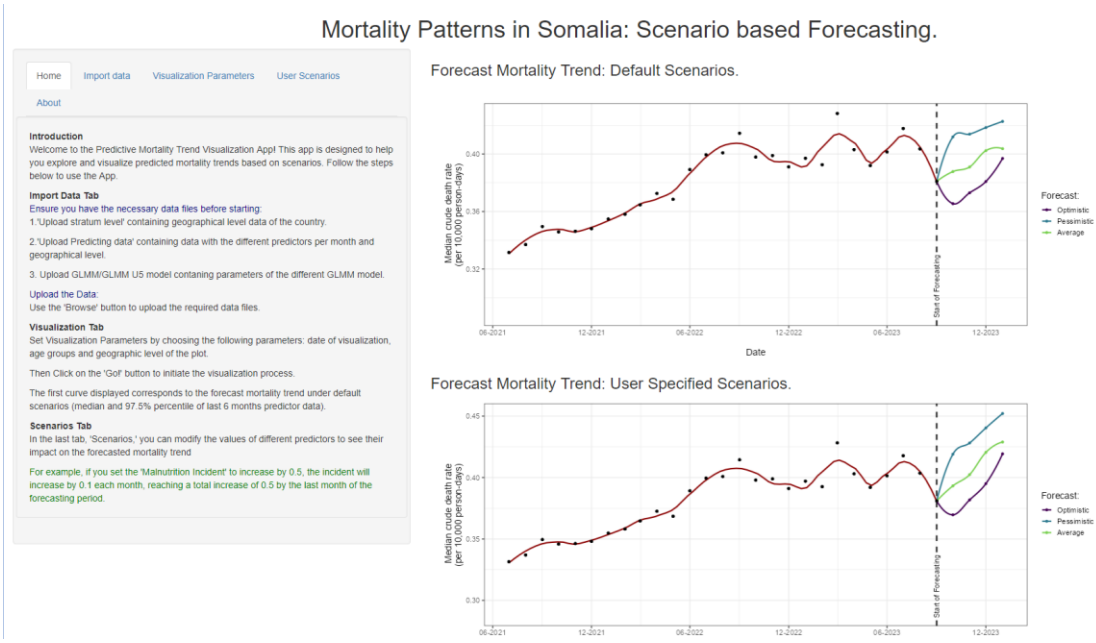

Figure 4. Screenshot of the RShiny App.

Here are the key details of the App:

Input Parameters

- **Admin2** *(Excel File):* Used to define the district and regional levels of the country, enabling the plotting of different curves at varying geographic levels.
- **Predicting Data** *(RDS File):* Contains all the different predictors measured at the district level, forming the foundation for scenario generation.
- **GLM Model** *(RDS File)*: Contains the coefficients of the Generalized Linear Mixed Model (GLMM) developed during the retrospective phase to predict crude death rates for the overall population.
- **GLM Model Under 5's** *(RDS File)*: Contains the coefficients of the GLMM model specifically designed for predicting crude death rates for individuals under 5 years old.

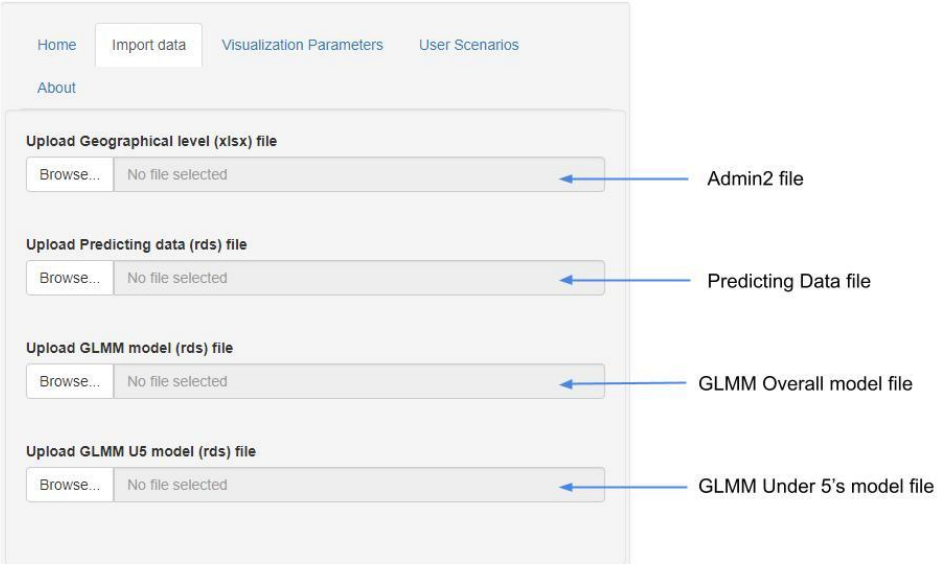

Figure 5. Screenshot of the Import Data Tab of the Rshiny App.

Specificities & Visualization

- **First Tab: Visualization Parameters**: Users have the flexibility to change various parameters for visualization:
  - Start Date of Visualization: Provides access to retrospective predictions, allowing users to specify the beginning of the visualization period.
  - Age Groups: Choose to plot curves for the overall population or for individuals under 5 years old.
  - Geographic Level: Choose to plot at the national or district level, suiting the visualization to different geographic levels.

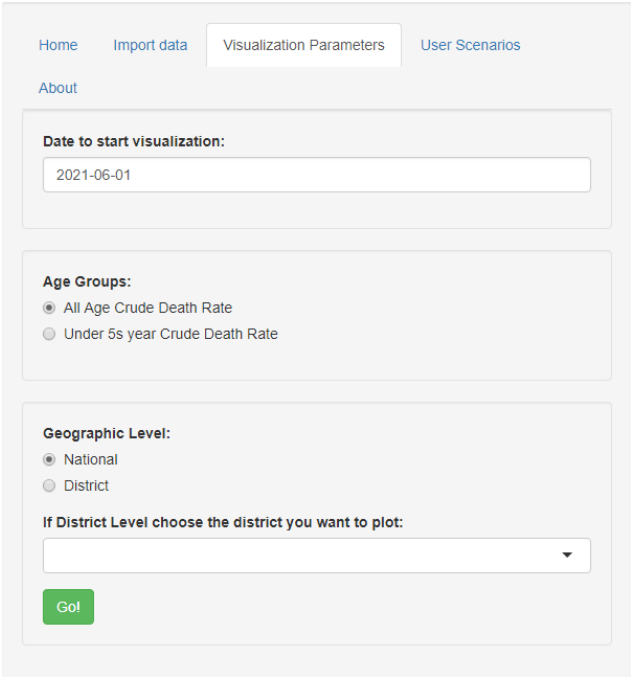

Figure 6. Screenshot of Visualization Parameters Tab of the Rshiny App.

- **Second tab: Predictor modification:** Users can select the predictor they wish to increase or decrease, allowing more detailed control over scenario customization.

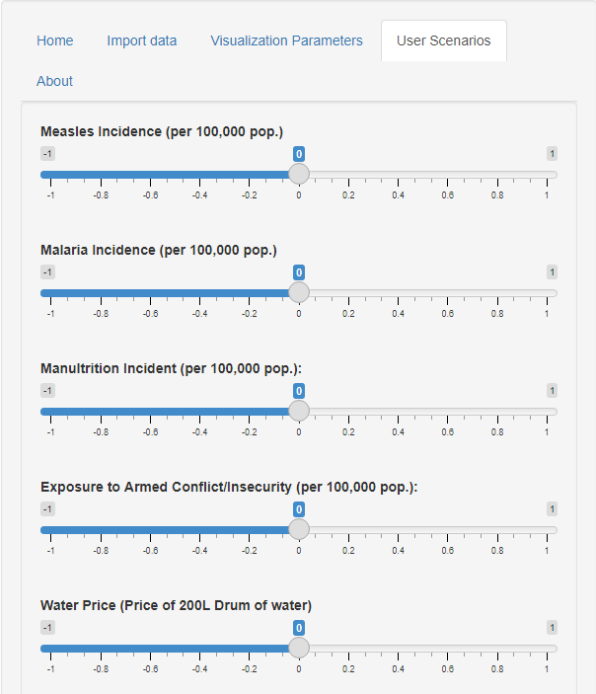

Figure 7. Screenshot of User Scenarios tab of the Rshiny App.

Results

To evaluate our methodology, we compare the results of crude mortality rates over a period when the values of the model's predictors are available. This allows us to derive predictions of crude mortality

rates using the GLMM model, contrasting them with forecast predictions using the default scenario methodology. Results are shown in Figure 8.

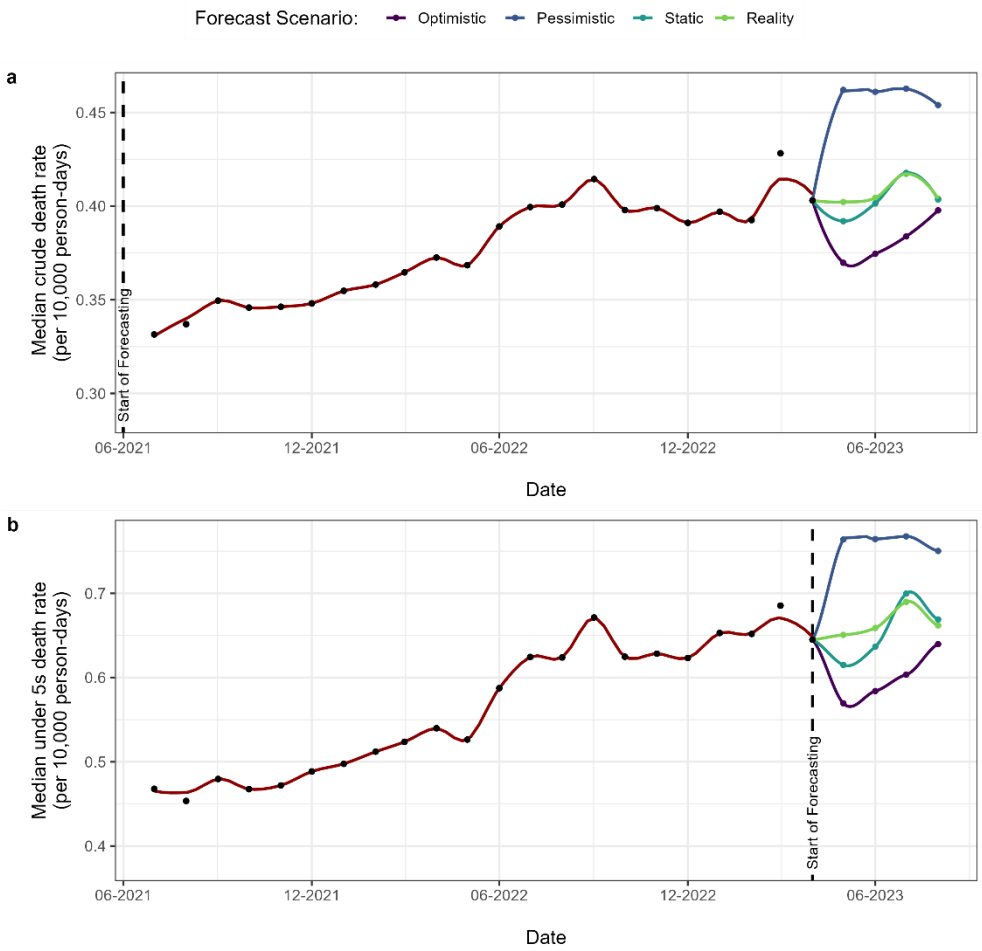

Figure 8. Forecast evolution of CDR (panel a) and U5DR (panel b) across Somalia over a time horizon of five months within a defined timeframe, aiming to assess the performance of our methodology. For visual clarity, only median model estimates are shown.

Limitations

The results seem reasonable, but there are some limitations due to the methodology used and the need to make the app useable by other users:

- **Simple Linear Correlation between Predictors**:
  - Issue: Relying on simple linear correlation assumes a linear relationship between predictors, which may not be accurate, especially when dealing with non-normally distributed data. Many predictors exhibit a 0-inflated distribution, assuming of normality questionable.
  - Possible solution: A better approach might involve exploring alternative methods such as Generalized Additive Models (GAM) or other machine learning models that can capture non-linear relationships more effectively.
- **Percentage of Increased or Decreased**:
  - Issue: Defining percentage modifications between -100% and 100% lacks specificity for

certain predictors. For instance, interpreting a decrease of -100% for a power index may not provide meaningful information.

  - Possible solution: It would be beneficial to suit the percentage range for each predictor, considering the specific characteristics and meaningful bounds for individual predictors.

- **Limited Access to All Predictor Relationships**:
  - Issue: The model might lack comprehensive coverage of all relationships between predictors. For example, incorporating variables like vaccine data could enhance the model's completeness and accuracy.
  - Possible solution: The model's predictive capabilities are constrained by the absence of certain predictor relationships. Access to additional relevant predictors could improve the overall model performance and reliability.

This app is a first stage to producing accessible forecasting tool. In time these limitations could be addressed to some degree. Exploring more sophisticated statistical methods, customizing percentage adjustments based on predictor characteristics. Some of these changes can be quick, but without prolonged use and feedback on how the tool is used, or what users want, this first step starts everything going.

References

1. R. J. Hyndman, G. Athanasopoulos, Forecasting: Principle and Practice, 2021, Otexts

2. E. B. Dagum, S. Bianconcini, Seasonal adjustment methods and real time trend-cycle estimation, 2016

3. W. Chang, J. Cheng, JJ Allaire, C. Sievert, B. Schloerke, Y. Xie, J. Allen, J. McPherson, A. Dipert, B. Borges, shiny: Web Application Framework for R, 2023

4. A. C. Aitken, Statistical Mathematics 8th Edition, 1957

5. M. E. Brooks, K. K. Kristensen, K. J. van Benthem, A. Magnusson, C. W. Berg, A. Nielsen, H. J. Skaug, M. Mächler, B. M. Bolker, glmmTMB Balances Speed and Flexibility Among Packages for Zero-inflated Generalized Linear Mixed Modeling, 2017